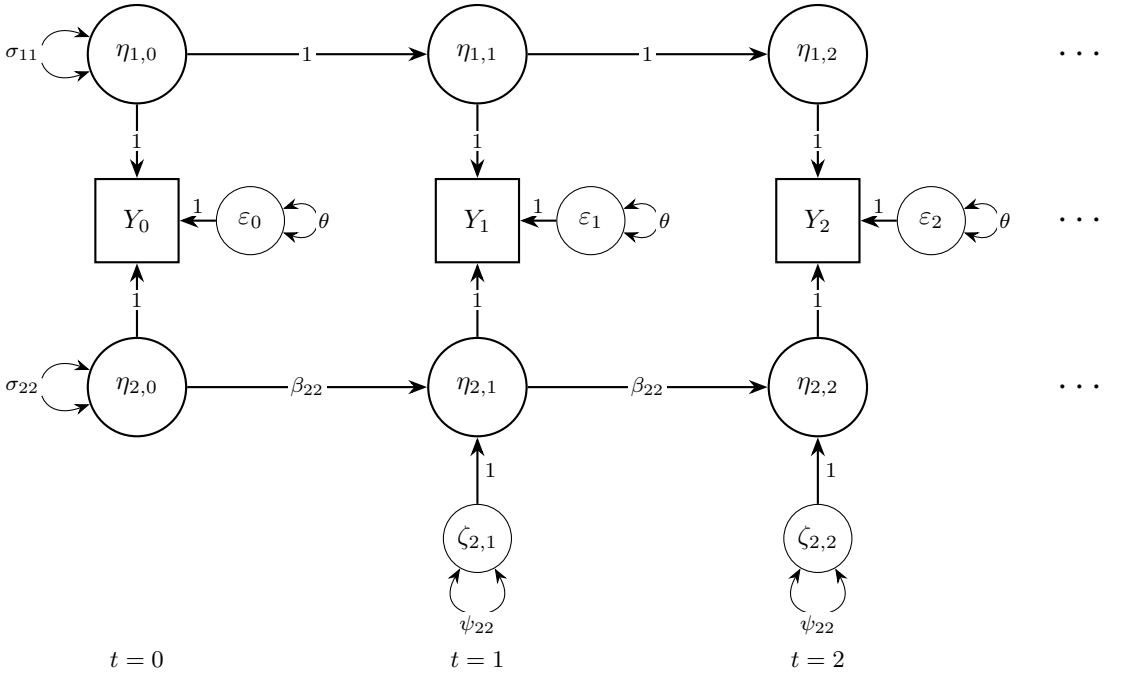


$$\boldsymbol{\eta}_t = (\eta_{1,t}, \eta_{2,t})', \quad \mathbf{y}_t = Y_t, \quad Y_t = \eta_{1,t} + \eta_{2,t} + \varepsilon_t$$

stable trait $\eta_{1,t} = \eta_1$ for all t; autoregressive trait $\eta_{2,t} = \beta_{22}\eta_{2,t-1} + \zeta_{2,t}$; the state is ε_t

$$\boldsymbol{\beta} = \begin{pmatrix} 1 & 0 \\ 0 & \beta_{22} \end{pmatrix} \qquad \boldsymbol{\Lambda} = \begin{pmatrix} 1 & 1 \end{pmatrix} \qquad \boldsymbol{\Psi} = \begin{pmatrix} 0 & 0 \\ 0 & \psi_{22} \end{pmatrix} \qquad \boldsymbol{\Theta} = \theta$$

$$\boldsymbol{\eta}_0 \sim \mathcal{N}(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0), \quad \boldsymbol{\mu}_0 = \begin{pmatrix} \mu_1 \\ 0 \end{pmatrix}, \quad \boldsymbol{\Sigma}_0 = \begin{pmatrix} \sigma_{11} & 0 \\ 0 & \sigma_{22} \end{pmatrix}$$